ACADEMIC PROJECT PROPOSAL

AI-Based Disaster Management & Rescue Support System

Software Engineering and Project Management • Capstone Proposal

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TECH STACK
Java 17, Java Swing GUI, MySQL 8.0, JDBC

PROCESS MODEL
Incremental Software Development Model

1. EXECUTIVE SUMMARY & OVERVIEW

The **AI-Based Disaster Management and Rescue Support System (ADMRSS)** is a centralized desktop management application engineered to optimize disaster response workflows. During natural hazard events like floods, cyclones, or urban fires, emergency response teams face extreme friction in distress call ingestion, urgency evaluation, field team deployment, and relief inventory logistics.

ADMRSS addresses these operational bottlenecks by unifying citizen emergency reporting, automated intelligent urgency triage, administrative dispatch control, and relief supply management into a cohesive, secure desktop system built with Java Swing and MySQL.

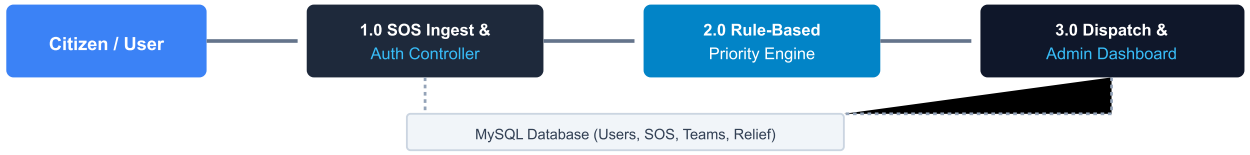
2. PROBLEM STATEMENT & FUNCTIONAL REQUIREMENTS

Traditional disaster response relies heavily on fragmented phone channels and manual call logs, creating critical dispatch latency and stock misallocations.

REQ ID	MODULE	FUNCTIONAL REQUIREMENT DESCRIPTION
FR-01	Authentication	Secure user registration and Role-Based Access Control (RBAC).
FR-02	SOS Ingestion	Submit emergency distress calls with location telemetry and victim counts.
FR-03	Priority Engine	Automatically calculate emergency urgency score using rule matrix.
FR-04	Rescue Dispatch	Register rescue units and assign available teams to active SOS incidents.
FR-05	Relief Control	Manage stock levels and record resource distributions (Food, Water, Medicine).
FR-06	Dashboard	Display real-time incident counters, team statuses, and inventory metrics.

3. SYSTEM ARCHITECTURE & HIGH-LEVEL DFD

The application adheres to a strict 3-Layer Architecture separating presentation GUI, service logic, and database operations.



4. DATABASE SCHEMA DESIGN (MYSQL)

TABLE NAME	PRIMARY KEY	FOREIGN KEYS	PURPOSE & KEY COLUMNS
`users`	user_id	-	User accounts, email, password_hash, role
`disasters`	disaster_id	-	Disaster events, type, location, severity
`sos_requests`	request_id	user_id, disaster_id	Emergency reports, priority, status, telemetry
`rescue_teams`	team_id	-	Field units, member counts, availability status
`rescue_assignments`	assignment_id	request_id, team_id	Active deployment logs, assigned timestamps
`relief_resources`	resource_id	-	Inventory tracking for food, water, and medicine

5. INTELLIGENT PRIORITY CALCULATION MATRIX

Rule-Based Decision Logic: The first version implements an algorithmic decision matrix to categorize urgency levels dynamically based on distress input parameters:

CRITICAL: LIFE DANGER = TRUE OR AFFECTED > 10

HIGH: TRAPPED = TRUE OR INJURED = TRUE

MEDIUM: FOOD/WATER DEPLETED = TRUE

6. CORE PRIORITY ENGINE IMPLEMENTATION (JAVA)

```
package com.admrss.service; import com.admrss.model.SOSRequest; public class PriorityCalc
```

7. PROJECT SCHEDULE & INCREMENTAL ROADMAP

PHASE	MILESTONE ACTIVITY	TARGET TIMELINE	DELIVERABLE
Inc 1	Requirement SRS & DB Schema Setup	Weeks 1 – 2	Approved SRS & ERD
Inc 2	JDBC Persistence & Auth Module	Weeks 3 – 4	Login / Register GUI
Inc 3	SOS Ingestion & Priority Calculator	Weeks 5 – 6	Triage Core System
Inc 4	Rescue Dispatch & Relief Module	Weeks 7 – 8	Dispatch & Inventory Panel
Inc 5	Dashboard Analytics & Integration	Weeks 9 – 10	Integrated Platform
Inc 6	System Testing & Final Release	Weeks 11 – 12	Final Report & Code Base

8. VERIFICATION & FUTURE SCOPE

The initial prototype has been verified for correct priority classification, role security, and real-time database state transitions. Future iterations will replace the rule-based triage system with a trained Machine Learning model and integrate GIS mapping APIs for spatial coordinate visualization.